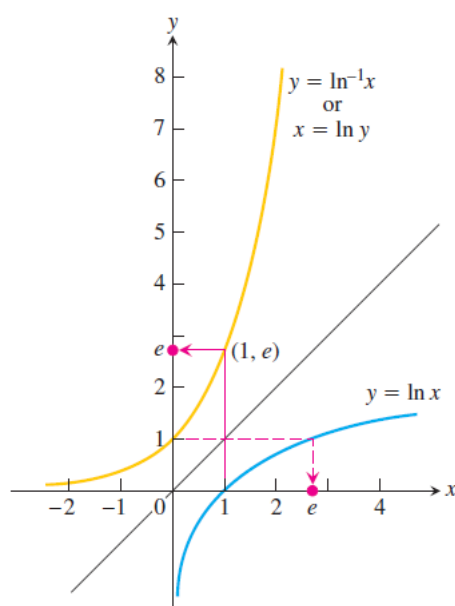


7.3 Exponential Functions

Having developed the theory of the function $\ln x$, we introduce its inverse, the exponential function $\exp x = e^x$. We study its properties and compute its derivative and integral.

① The Inverse of $\ln x$

The function $\ln x$, being an increasing function of x with domain $(0, \infty)$ and range $(-\infty, \infty)$, has an inverse $\ln^{-1} x$ with domain $(-\infty, \infty)$ and range $(0, \infty)$. The graph of $\ln^{-1} x$ is the graph of $\ln x$ reflected across the line $y = x$.



The function $\ln^{-1} x$ is usually denoted as $\exp x$. We can show that $\exp x$ is an exponential function with base e .

DEFINITION For every real number x , we define the **natural exponential function** to be $e^x = \exp x$.

② Inverse Equations for e^x and $\ln x$

$$e^{\ln x} = x \quad (\text{all } x > 0)$$

$$\ln e^x = x \quad (\text{all } x)$$

Example 1 Solve the equation $e^{2x-6} = 4$ for x .

Example 2 Find k if $e^{2k} = 10$.

③ The Derivative and Integral of e^x

◆ If u is any differentiable function of x , then

$$\frac{d}{dx} e^x = e^x, \quad \frac{d}{dx} e^u = e^u \frac{du}{dx}.$$

Example 3 Find derivatives of the exponential.

$$(a) \frac{d}{dx}(5e^x); \quad (b) \frac{d}{dx}(e^{-x}); \quad (c) \frac{d}{dx}(e^{\sin x}); \quad (d) \frac{d}{dx}(e^{\sqrt{3x+1}}); \quad (e) \frac{d}{dx}(x^2 e^{2x}).$$

◆ The general antiderivative of the exponential function

$$\int e^u du = e^u + C.$$

Example 4 Evaluate

$$(a) \int_0^{\ln 2} e^{3x} dx; \quad (b) \int_0^{\frac{\pi}{2}} e^{\sin x} \cos x \, dx.$$

④ Laws of Exponents

THEOREM 3 For all numbers x , x_1 , and x_2 , the natural exponential e^x obeys the following laws:

$$(a) e^{x_1} e^{x_2} = e^{x_1+x_2} \qquad (b) e^{-x} = \frac{1}{e^x}$$

$$(c) \frac{e^{x_1}}{e^{x_2}} = e^{x_1 - x_2}$$

$$(d) (e^{x_1})^r = e^{rx_1}, \text{ if } r \text{ is rational}$$

⑤ The General Exponential Function a^x

Since $a = e^{\ln a}$ for any positive number a , we can express a^x as $(e^{\ln a})^x = e^{x \ln a}$. We therefore use the function e^x to define the other exponential functions, which allow us to raise *any* positive number to an irrational exponent.

DEFINITION For any numbers $a > 0$ and x , the **exponential function with base a** is

$$a^x = e^{x \ln a}.$$

Hence, we get

$$(a^x)' = (e^{x \ln a})' = e^{x \ln a} \cdot \ln a = a^x \ln a$$

$$(a^u)' = a^u \ln a \frac{du}{dx}$$

and the integral equivalent of this last result gives the general antiderivative

$$\int a^u du = \frac{a^u}{\ln a} + C.$$

Example 5 Find derivatives and integrals

$$(a) (3^x)'$$

$$(b) (3^{-x})'$$

$$(c) (3^{\sin x})'$$

$$(d) \int 3^x dx$$

$$(e) \int 2^{\sin x} \cos x dx$$

Example 6 Find y' if $y = x^x$, $x > 0$.

⑥ Logarithms with Base a

If a is any positive number other than 1, the function a^x is one-to-one and has a nonzero derivative at every point. It therefore has a differentiable inverse. We call the inverse the **logarithm of x with base a** and denote it by **$\log_a x$** .

DEFINITION For any positive number $a \neq 1$,

$$\log_a x$$

is the inverse function of a^x .

$$\text{So, } y = \log_a x \Leftrightarrow a^y = x.$$

⑦ Inverse Equations for a^x and $\log_a x$

$$\begin{aligned} a^{\log_a x} &= x & (x > 0) \\ \log_a(a^x) &= x & (\text{all } x) \end{aligned}$$

Example 7

$$(a) \log_2(2^5) = 5; \quad (b) \log_{10}(10^{-7}) = -7; \quad (c) 2^{\log_2 3} = 3; \quad (d) 10^{\log_{10} 4} = 4.$$

⑧ Derivatives of $\log_a x$

Note that $\log_a x = \frac{\ln x}{\ln a}$. Hence, we get

$$\begin{aligned} (\log_a x)' &= \frac{1}{\ln a} \frac{1}{x} \\ (\log_a u)' &= \frac{1}{\ln a} \frac{1}{u} u' \end{aligned}$$

Example 8 Find the derivative of

- (a) $y = \log_{10}(3x+1)$
- (b) $y = \log_4(2x^2+1)$
- (c) $y = \log_2(x^3+2x+1)$

⑨ The Number e Expressed as a Limit

$$e = \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x.$$